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INTERNATIONAL RELATIONS

OCPSG Research Programme

From Bronze to Benchmark: Validating Parliamentary Speech Labels in Spanish for Policy Agenda Research

Bastián González-Bustamante, Christopher Klamm, and Tom Bellens

<https://ocpsg-benchmarking-llms.github.io>

Presentation at the XVIII Spanish Congress of Political Science, Granada, Spain, September 9-11, 2026



OCPSG Benchmarking LLMs

Bastián González-Bustamante, Tom Bellens, Christopher Klammer, and Marta Koch

Project Overview & Motivation

Substantive Goal

Comparative political analysis increasingly requires scalable approaches to classify parliamentary interventions into policy topics across languages and institutional settings.

Complex Classification

Assigning one of 21 major policy topics to parliamentary speeches.

Methodological Goal

Recent LLM advances have raised new questions about robustness, comparability, interpretability, computational cost, and whether prompt-based systems can outperform supervised classifiers trained for this task.

The project treats multilingual topic annotation not merely as a prediction problem, but as a **data quality problem** in computational social science.

Why Benchmarking Matters

Researchers increasingly use LLMs and fine-tuned Transformers to annotate political text at scale.

But this promise depends on a prior question: **Which models perform reliably, under which language conditions, and for which policy labels?**

The Risk

Misclassifications become measurement errors in downstream political analysis.

The Problem

Human gold-standard labels are expensive to produce across many countries and languages.

The Gap

When established classifiers disagree, no single output can be treated as ground truth.

Background

The Comparative Agendas Project



A speech on renewable energy, for example, may invoke environmental protection, industrial policy, employment, and fiscal instruments at the same time. Human coders can resolve these cases through training and adjudication, however, large multilingual corpora make such procedures costly.



González-Bustamante, Klamm, Bellens, and Koch (2026)

Speeches & Bills

Policy-relevant texts assigned to a common hierarchical codebook of major topics and subtopics.

Cross-National Scope

Enables comparison of policy attention over time across countries.

Hierarchical Codebook

Major topics and subtopics covering the full range of government activity.

Related Work

From Dictionaries to Transformers

Classical Methods

Dictionaries, topic models, Random Forest. Improved efficiency but struggled with complex categories and cross-domain transfer.

LLMs

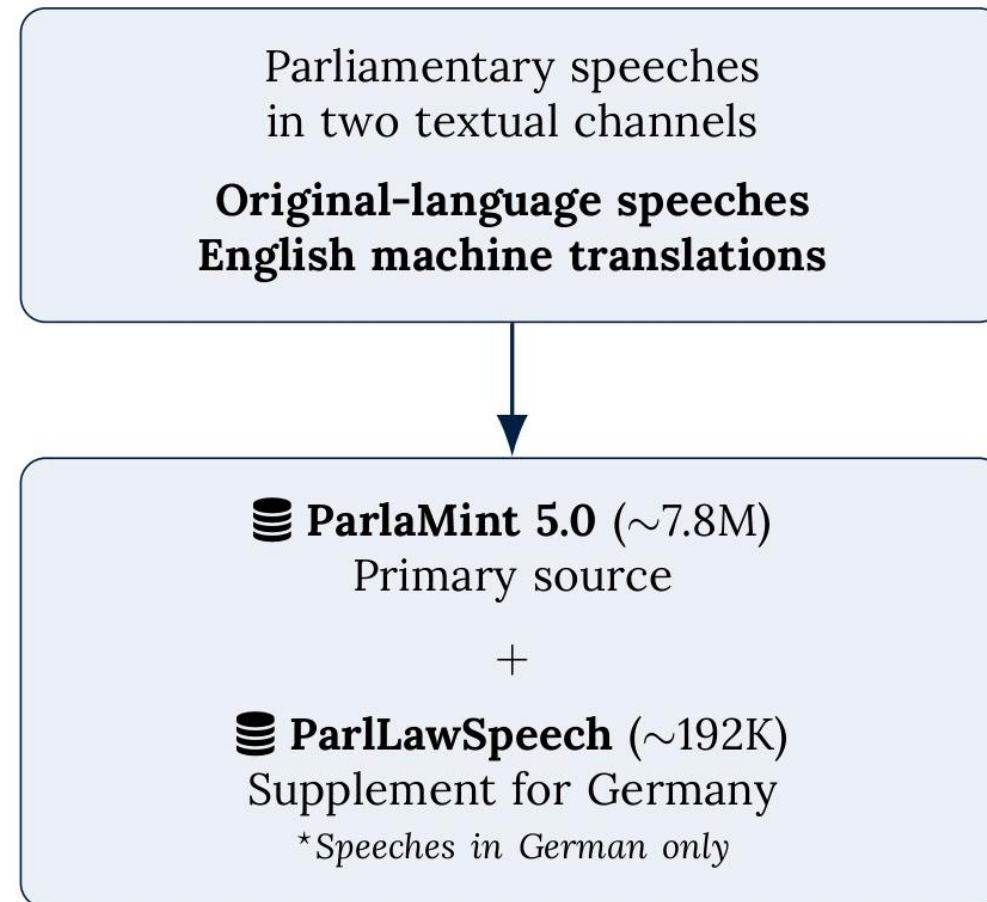
Zero-shot and few-shot annotation without training data, but performance depends on prompts, language coverage, and model updates.

Transformer Models

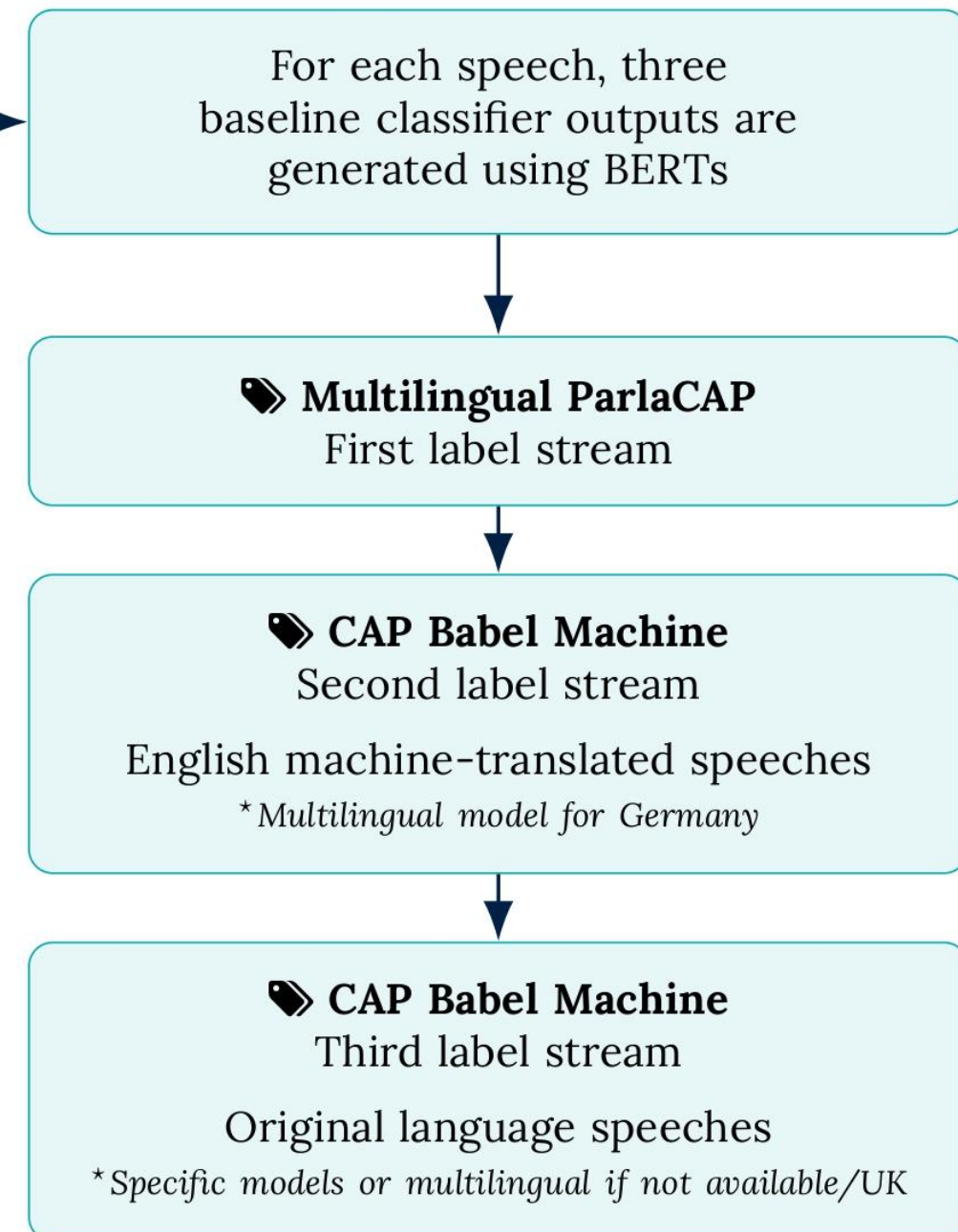
XLM-RoBERTa-based models improved policy agenda classification across languages, but results still vary by language, domain, and label.

❏ Performance varies across task design, language conditions, and model architectures. There is no universal solution.

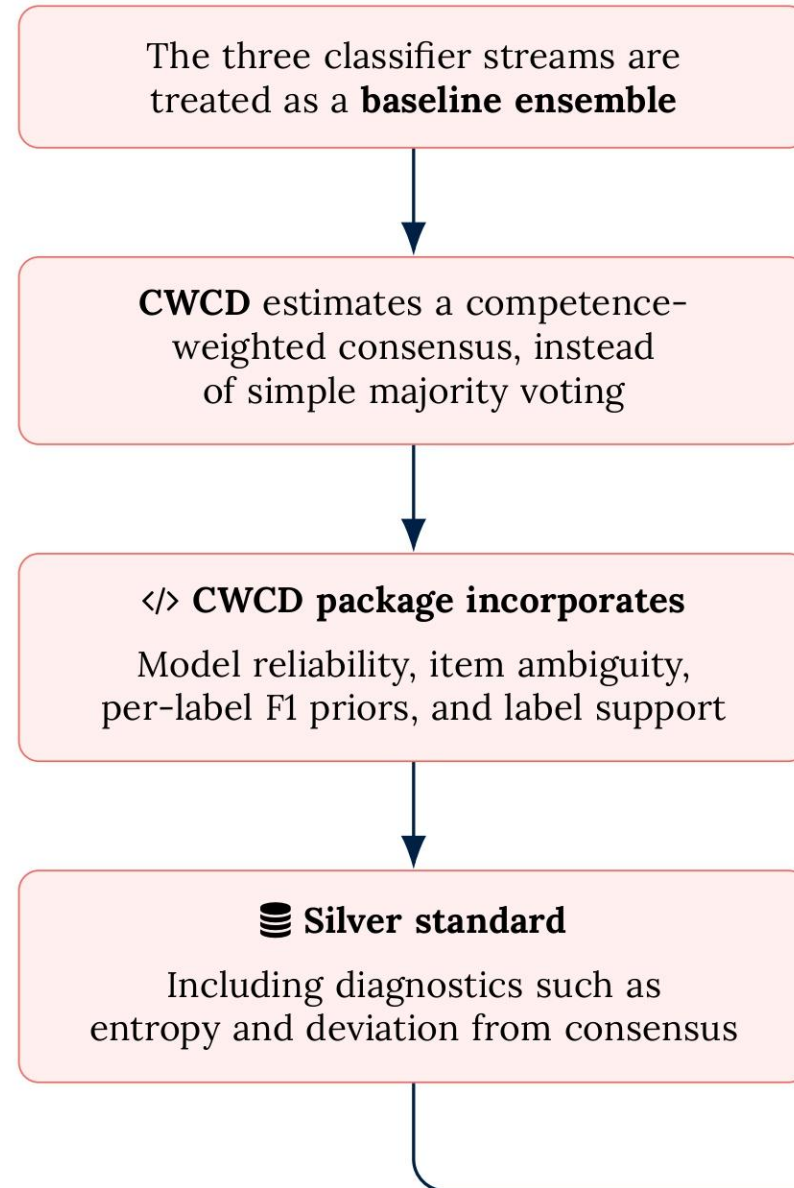
1. Corpus construction



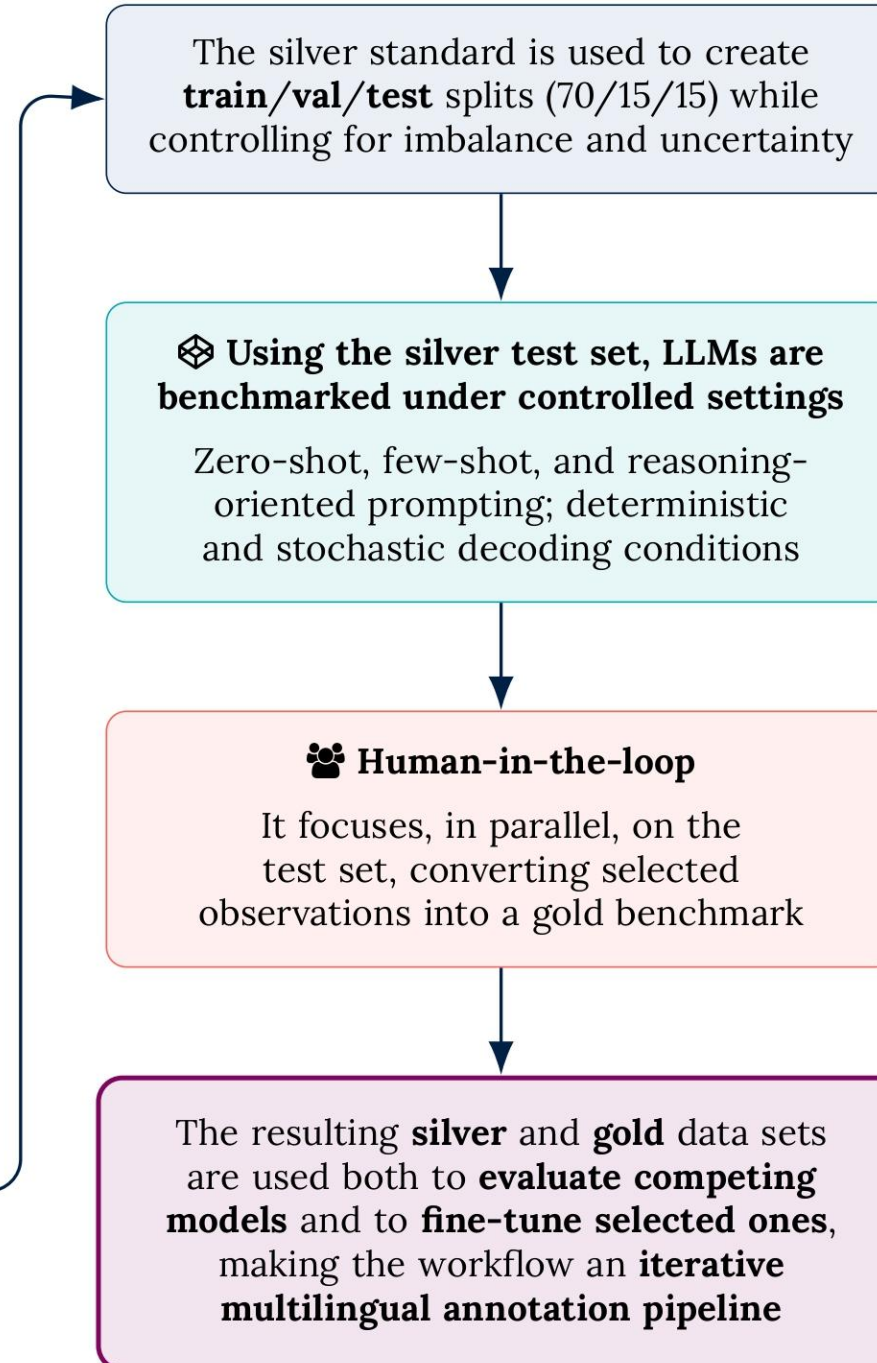
2. Baseline ensemble XLM-RoBERTa-large



3. Competence-weighted consensus



4. Split design and benchmarking



Three Baseline Classifiers

Stream 1: ParlaCAP

Existing topic-classification layer already linked to the parliamentary corpus. Provides multilingual labels directly comparable against other models.

Stream 3: CAP Babel Machine (original)

Same framework deployed on source-language speeches. Language-specific XLM-RoBERTa models used where available; multilingual variant otherwise.

Stream 2: CAP Babel Machine (MT)

Fine-tuned XLM-RoBERTa-large applied to English machine-translated speeches. Reduces complexity from language heterogeneity via a common channel.

** For Germany, ParlaCAP and CAP Babel Machine in both multilingual and German variants were used.*

Coverage across
countries in the
dataset.

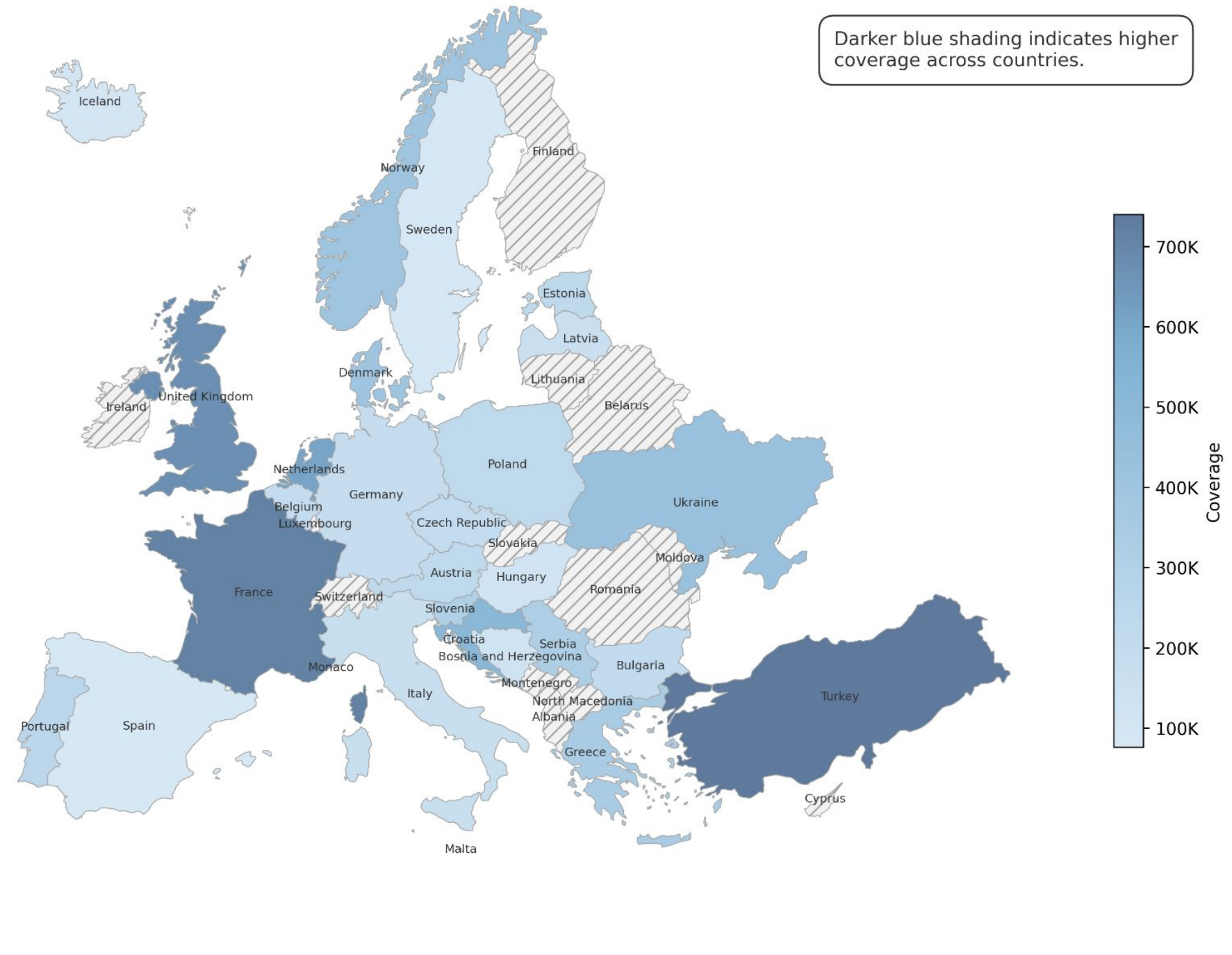
Values indicate the
number of speeches
or observations
included per country.

Data correspond to
ParlaMint 5.0,
except for Germany,
which corresponds to
ParLawSpeech.

No data are shown with
hatched shading.

Average: 307.8K

Darker blue shading indicates higher
coverage across countries.



Agreement between
ParlaCap and Babel
Machine across
countries.

Values are reported
as proportions.

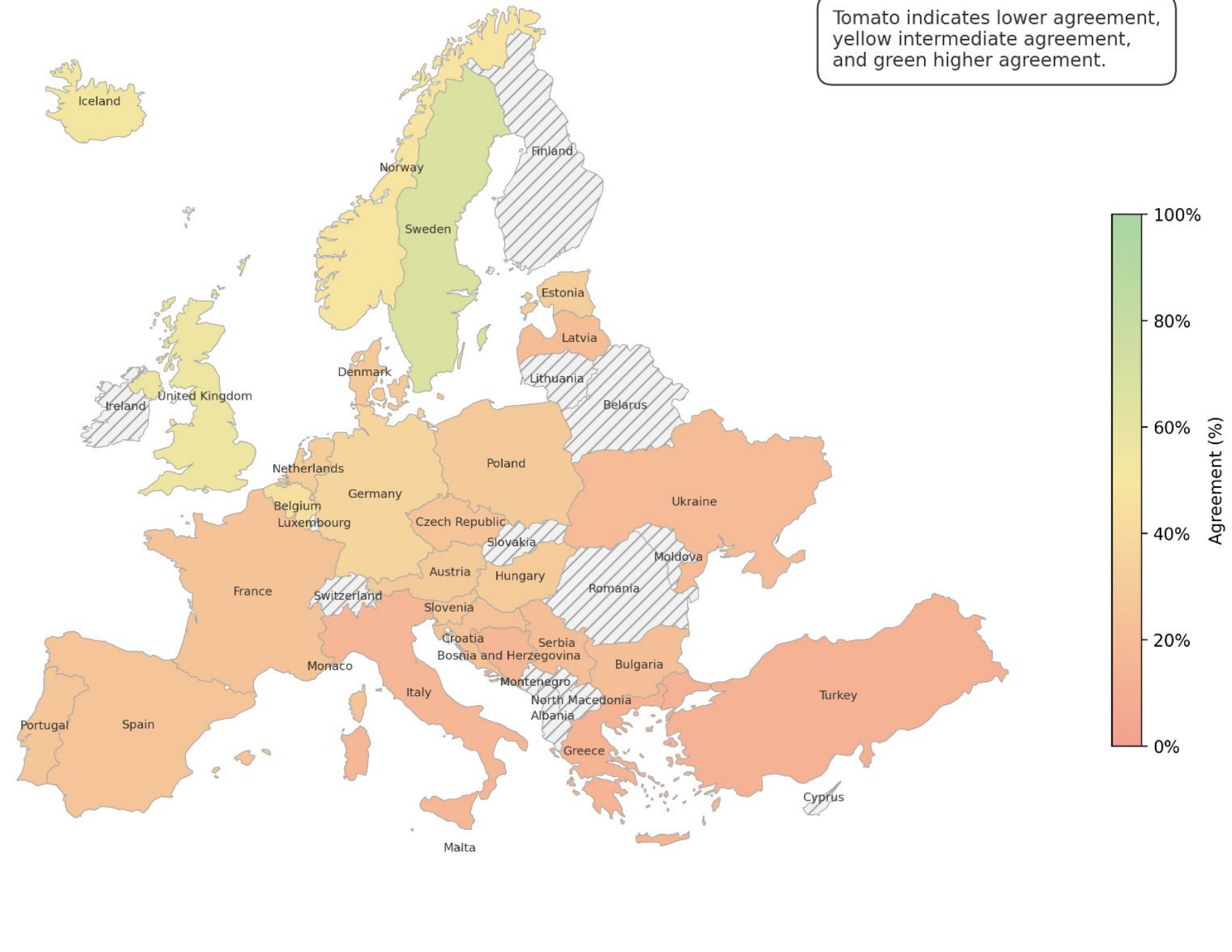
ParlaCap is an
extension of
ParlaMint that
includes automatically
annotated policy labels.

We ran Babel Machine
(XLM-RoBERTa-large)
on machine-translated
speeches.

No data are shown with
hatched shading.

Average: 30.5%

Tomato indicates lower agreement,
yellow intermediate agreement,
and green higher agreement.



Agreement across
three classifiers
by country.

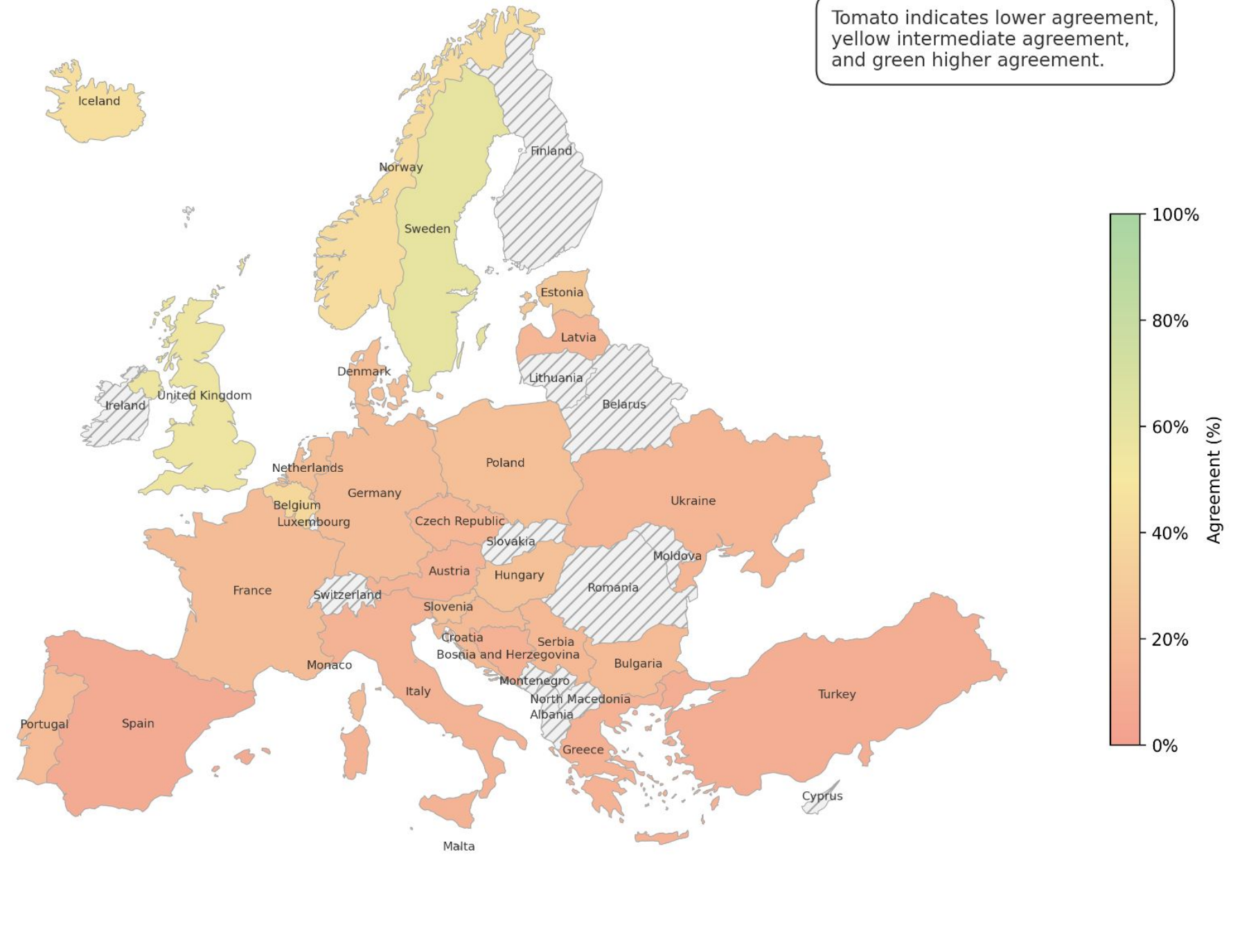
Values are reported
as proportions.

This measure extends
the previous
comparison by
incorporating an
additional classification
using original-language
speeches.

No data are shown with
hatched shading.

Average: 23.8%

Tomato indicates lower agreement,
yellow intermediate agreement,
and green higher agreement.



The Core Problem

Why Simple Agreement Fails

Majority Vote

Assumes equal expertise; forces hard labels regardless of confidence.

Dawid-Skene & MACE

Treats the true label as latent, estimates annotator-specific error patterns, models ability, bias, and item difficulty.

When classifiers disagree...

Their disagreement reveals uncertainty in the labelling process. A simple majority vote treats all classifiers as equally informative and forces a hard label even when signals are weak or contradictory. Model-based approaches offer a better foundation.

CWCD: Competence-Weighted Consensus

A central methodological innovation. Rather than merging classifier outputs by naïve majority vote, the project aggregates them through **CWCD (beta package)**, developed from the logic of MACE, adapted for probability-aware model consensus.



Model Competence

Estimates which classifier is most reliable, on which labels, and to what degree using Variational Bayes EM.



Item Ambiguity

Identifies which observations are intrinsically ambiguous via consensus entropy.



Model Deviation

Measures how far each classifier departs from the inferred consensus using competence-penalised Jensen-Shannon divergence.

The CWCD Framework

From Hard Labels to Soft Consensus

CWCD extends MACE from hard-label aggregation to soft-label consensus estimation. Each speech becomes an item with three model-generated annotations. CWCD estimates a latent consensus distribution over labels, not a simple arithmetic reconciliation.

F1 Priors & Support Weighting

Per-label F1-scores and label support from the test set are incorporated as priors. This recognises that a classifier may be strong on some policy topics and weak on others. **Support-based shrinkage** ensures labels estimated on very small samples do not disproportionately influence the consensus.

- Agreement among three streams differing in model origin and language channel is more informative than repeated predictions from a single pipeline.

The CWCD Framework

CWCD treats automated classifiers as noisy annotators. For each speech, it estimates:

- **Classifier competence.** The probability that a classifier's output is informed by the text rather than habitual label distributions.
- **Consensus label.** The mode of the posterior over the latent true label.
- **Item-level entropy.** Measuring residual uncertainty in the consensus distribution.

CWCD improves on majority voting by not assuming equal competence, using soft probability distributions, and producing an uncertainty estimate for every item.

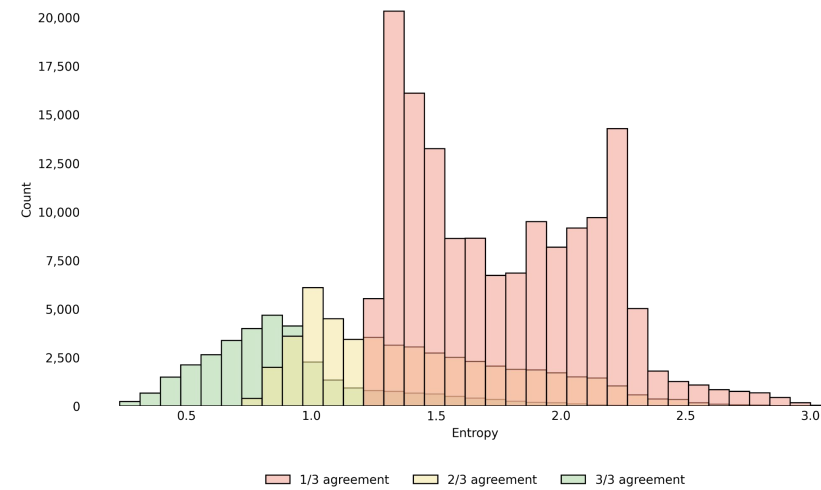
The consensus distribution and entropy for each speech are defined as:

$$c_{d,l} = \frac{\sum_a \theta_a q_{d,a,l}}{\sum_a \theta_a}, \quad H_d = - \sum_l c_{d,l} \ln c_{d,l}$$

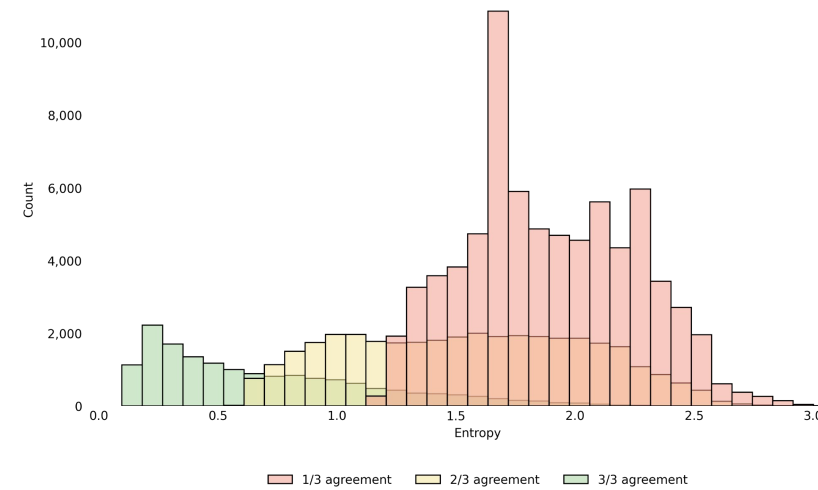
Where $q_{\{d,a,l\}}$ is classifier a 's probability for label l on speech d , and $\theta_{\{a\}}$ is classifier competence.

Low entropy indicates concentrated, reliable signals; high entropy indicates substantial ambiguity. Parameters are estimated by Variational Bayes EM with multiple random restarts, initialised from held-out per-label F1 scores.

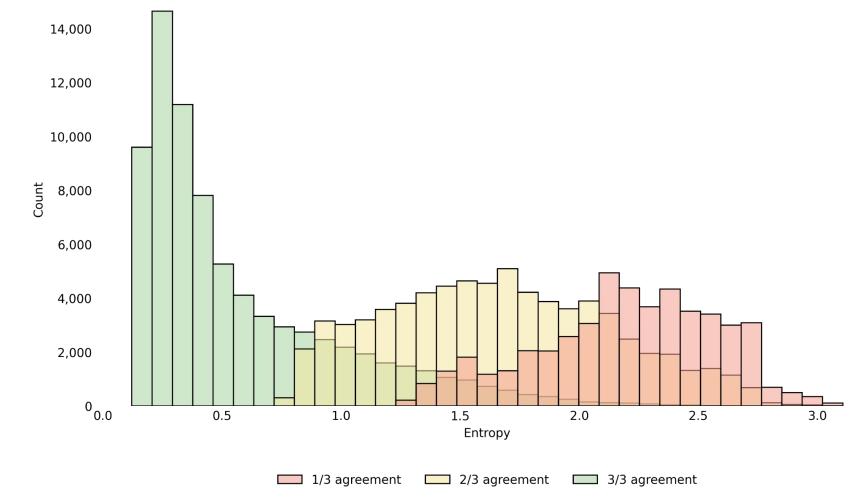
Entropy Distribution



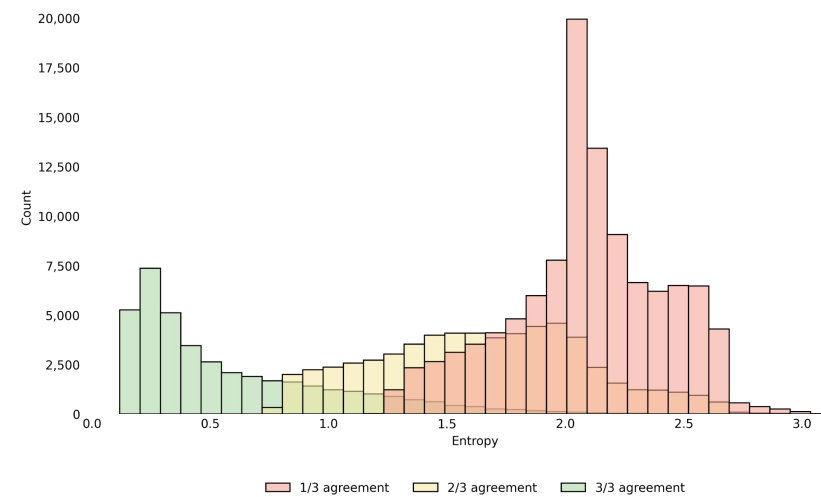
Austria



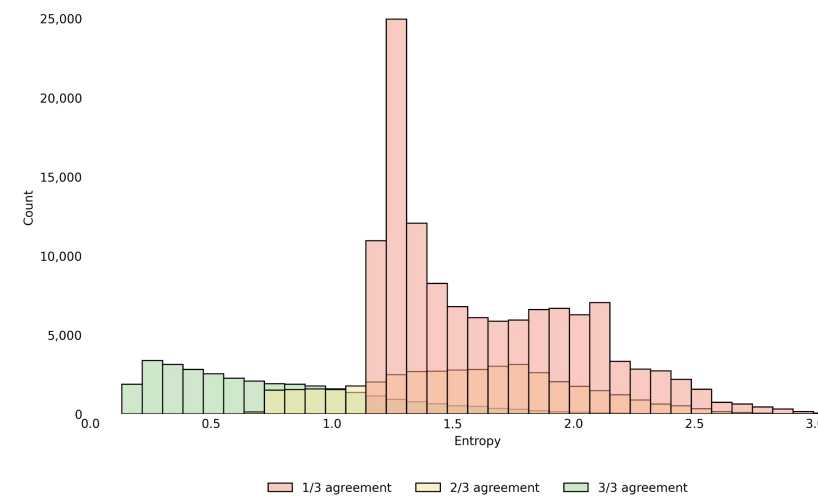
Bosnia and Herz.



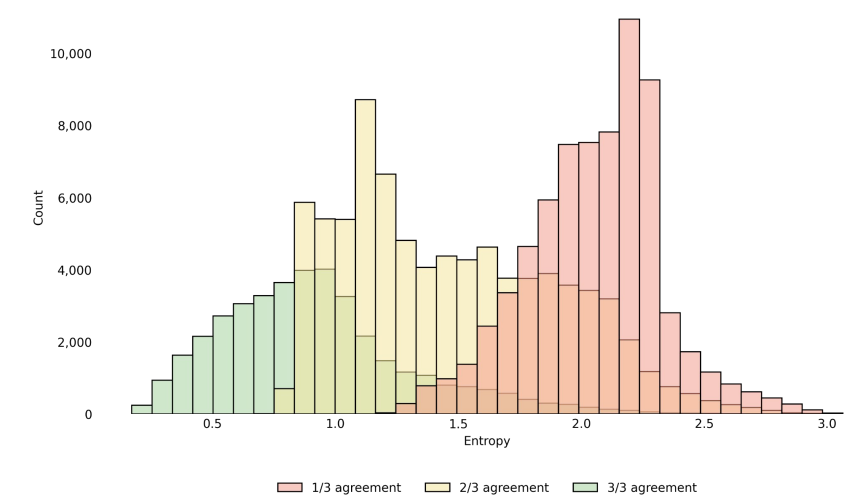
Belgium



Bulgaria

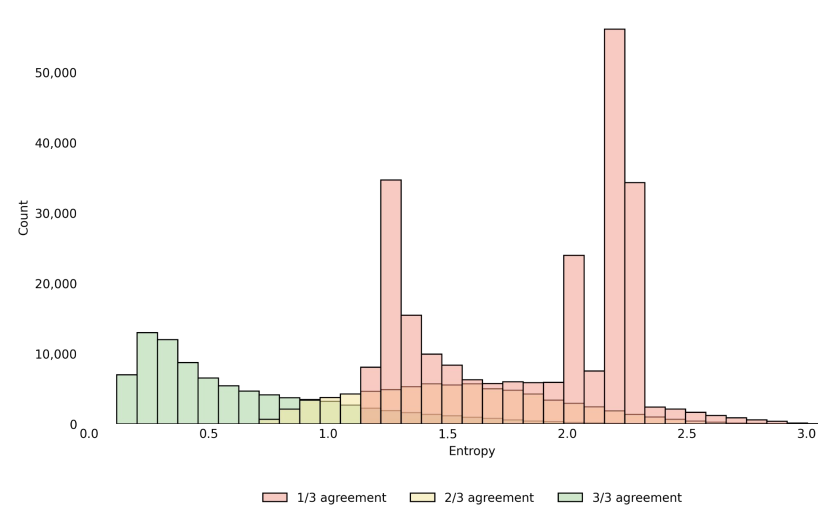


Czech Republic

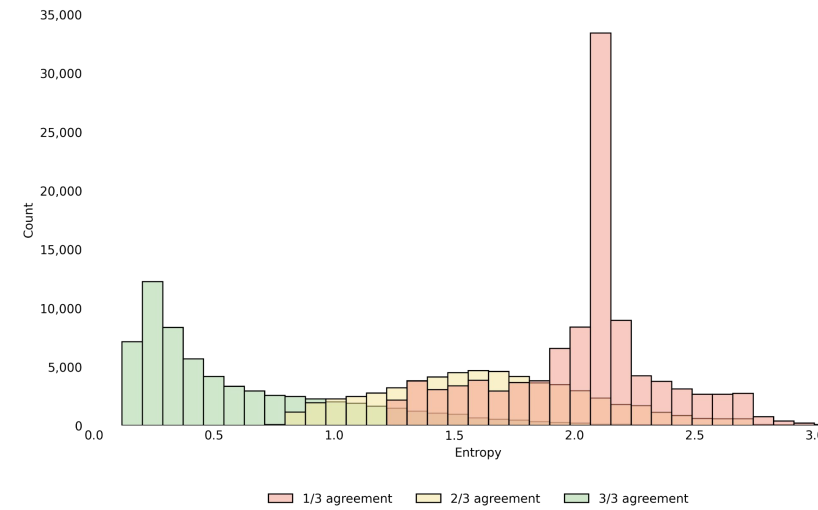


Germany

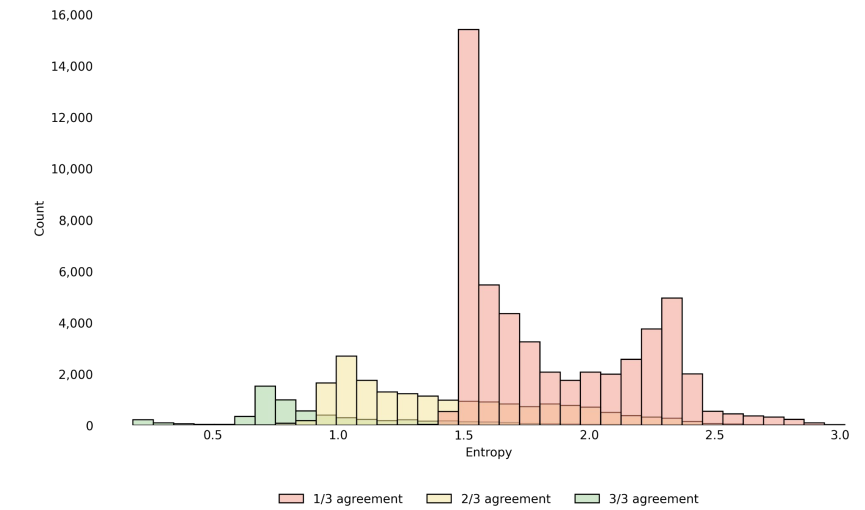
Entropy Distribution



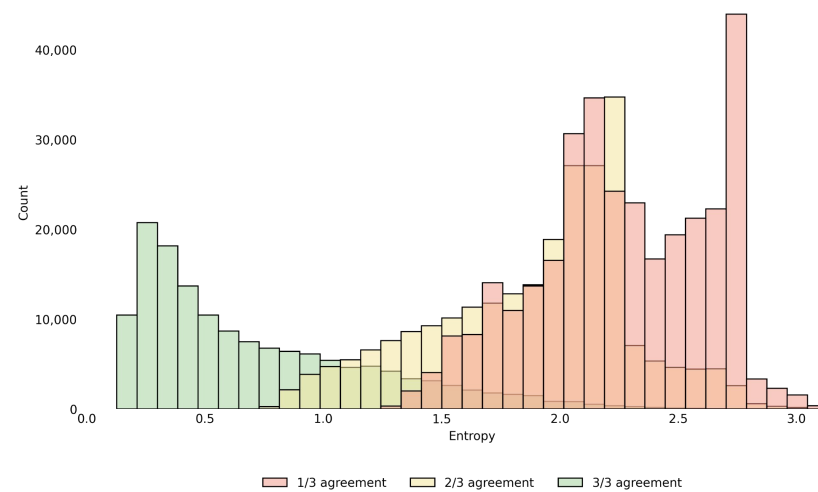
Denmark



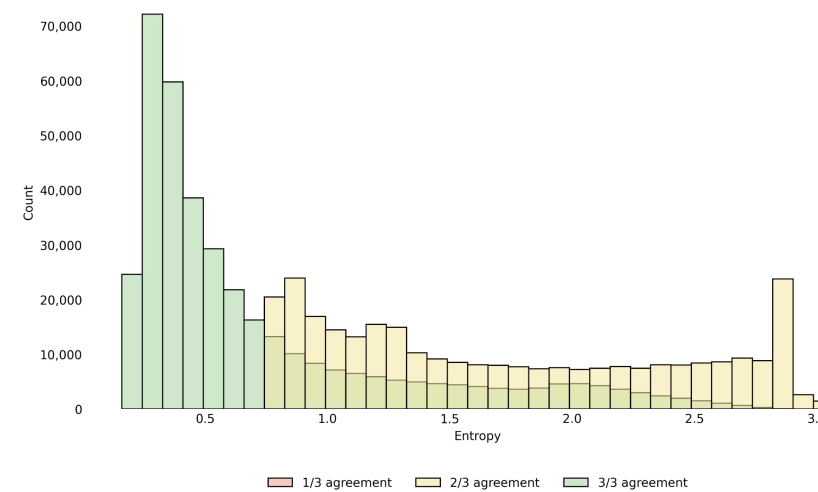
Estonia



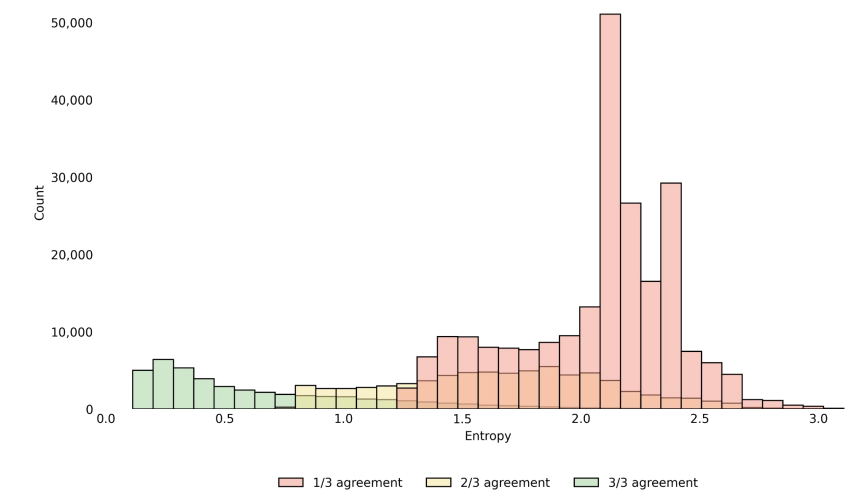
Spain



France

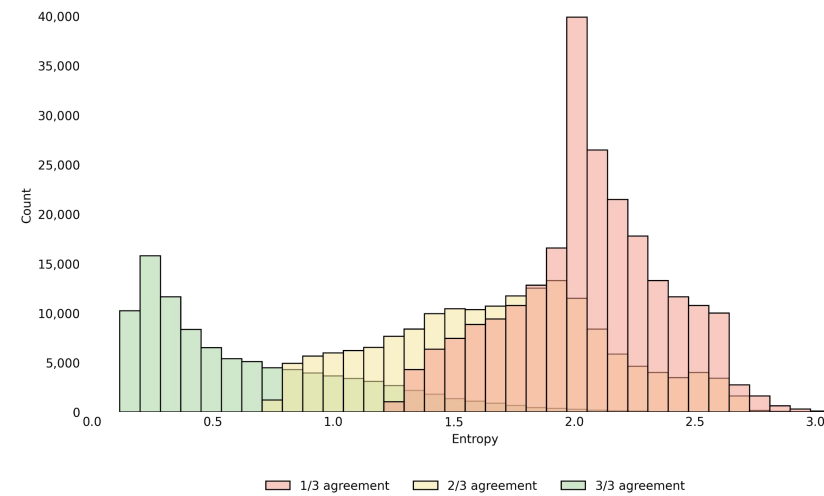


Great Britain

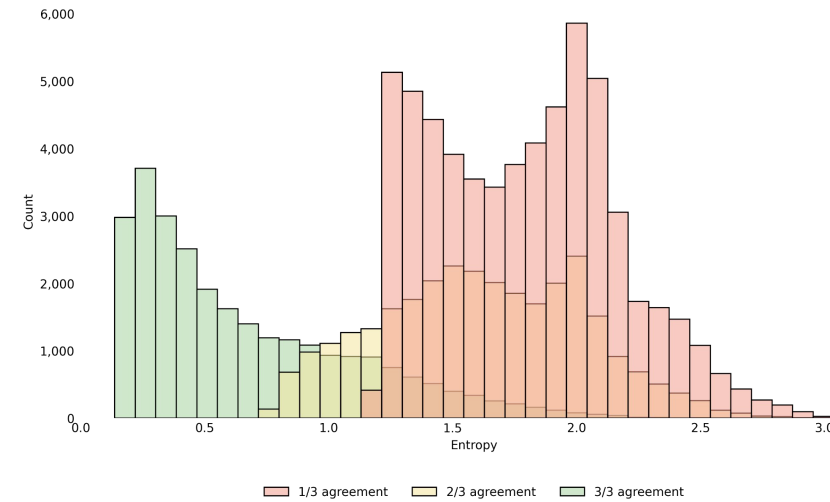


Greece

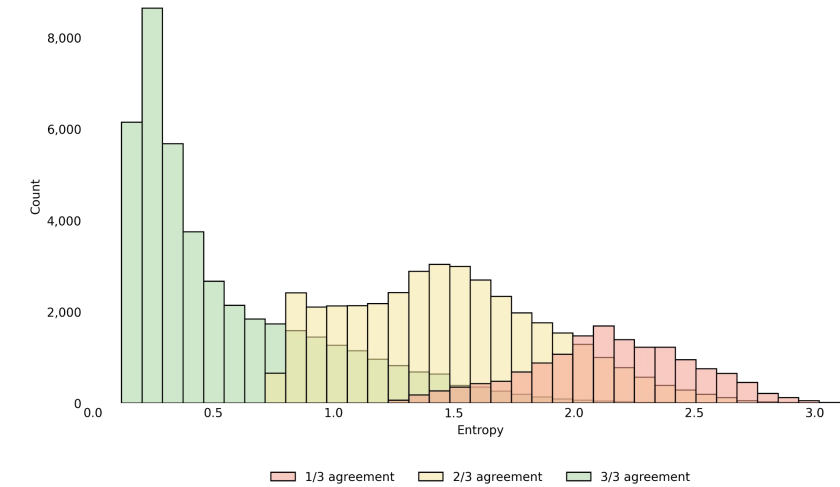
Entropy Distribution



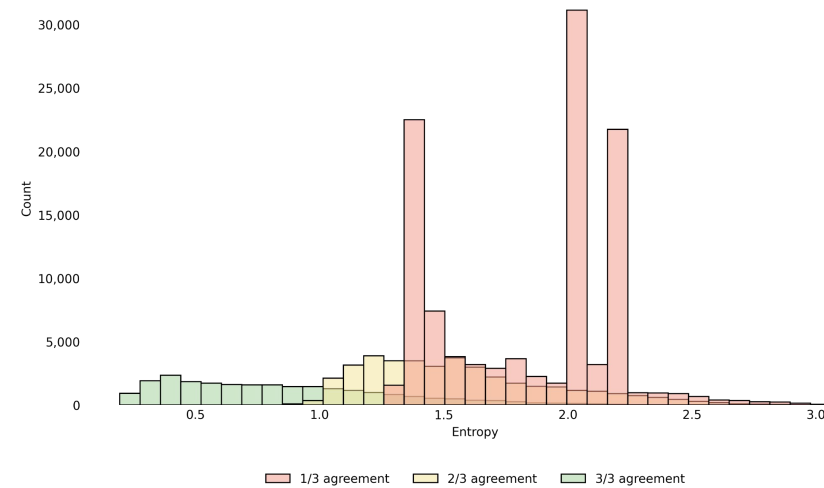
Croatia



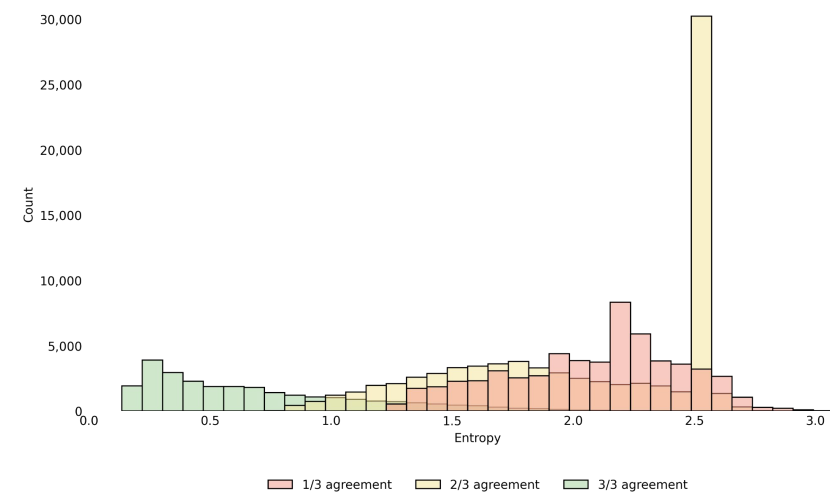
Hungary



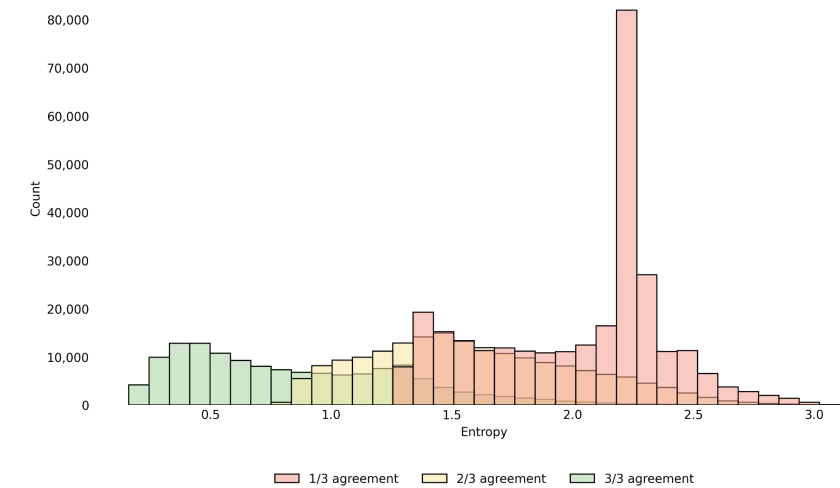
Iceland



Italy

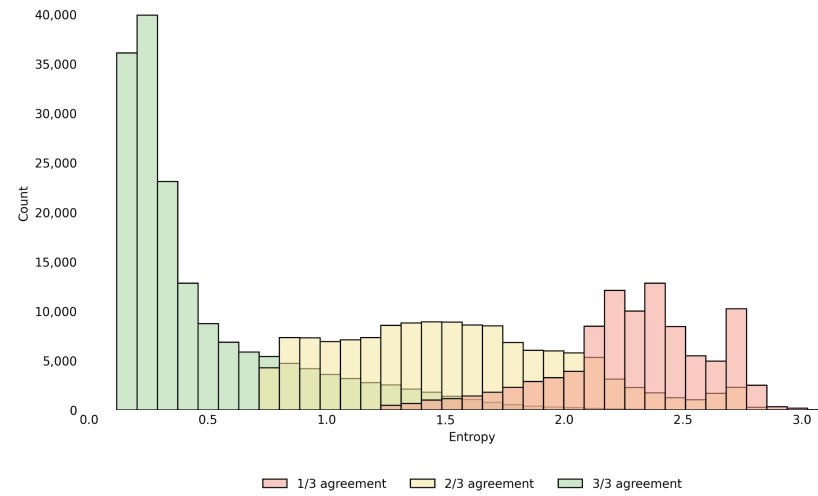


Latvia

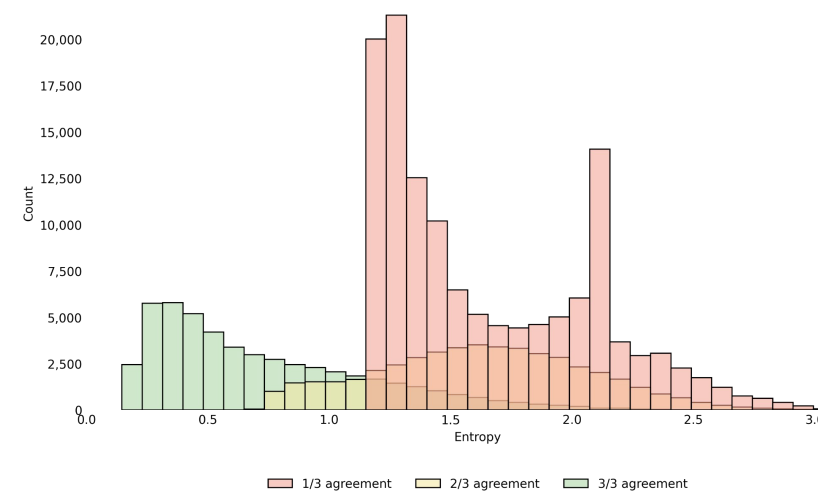


Netherlands

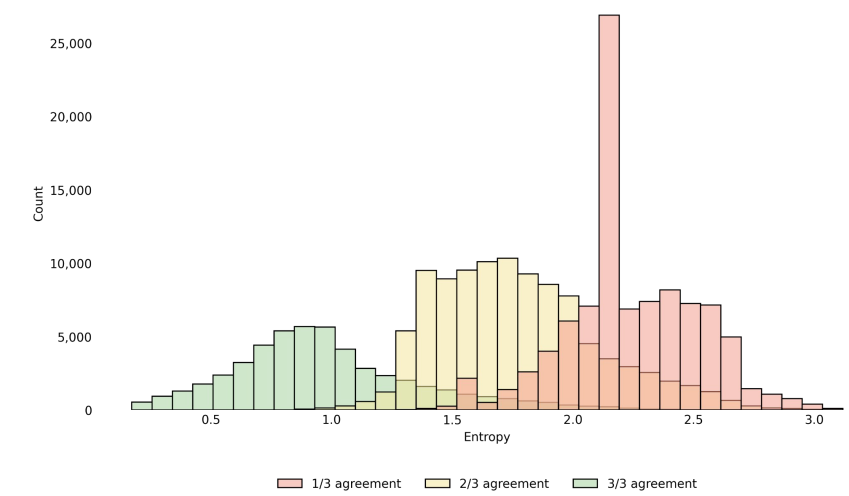
Entropy Distribution



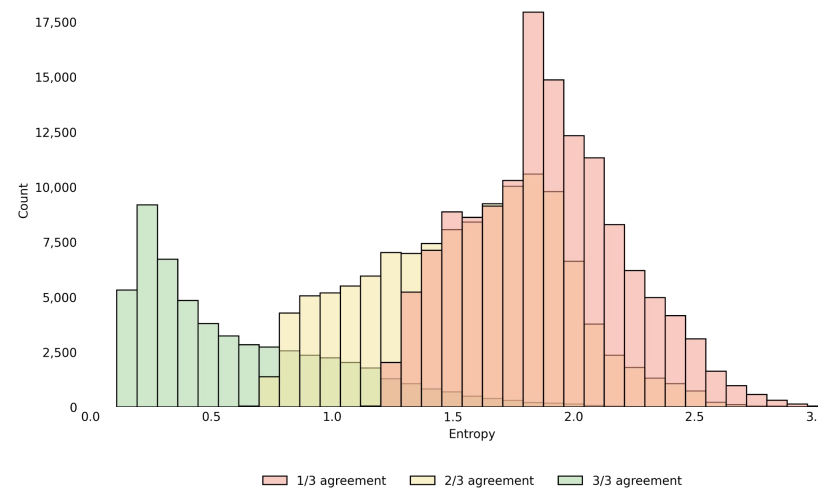
Norway



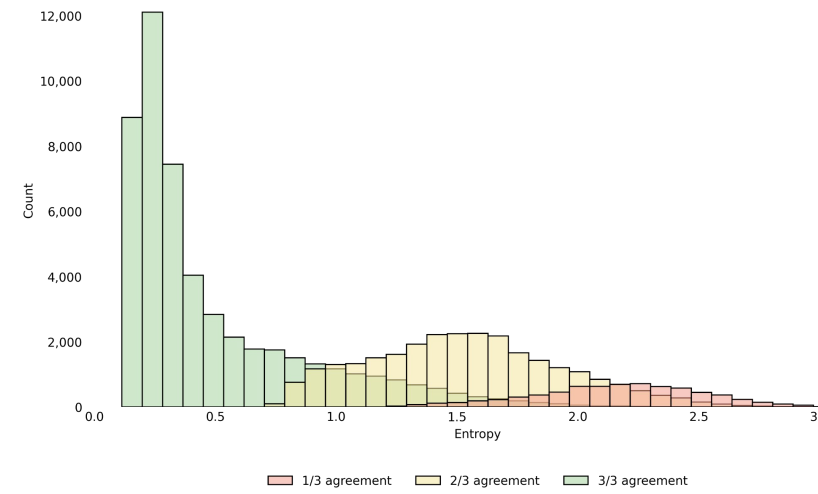
Poland



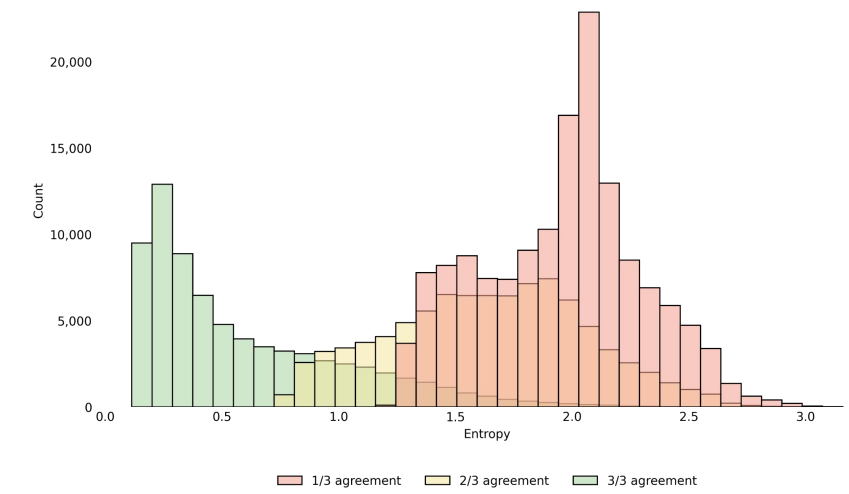
Portugal



Serbia

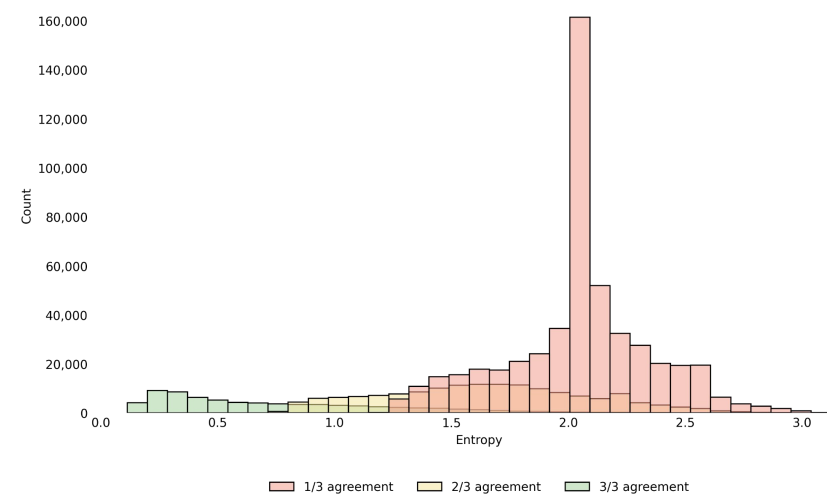


Sweden

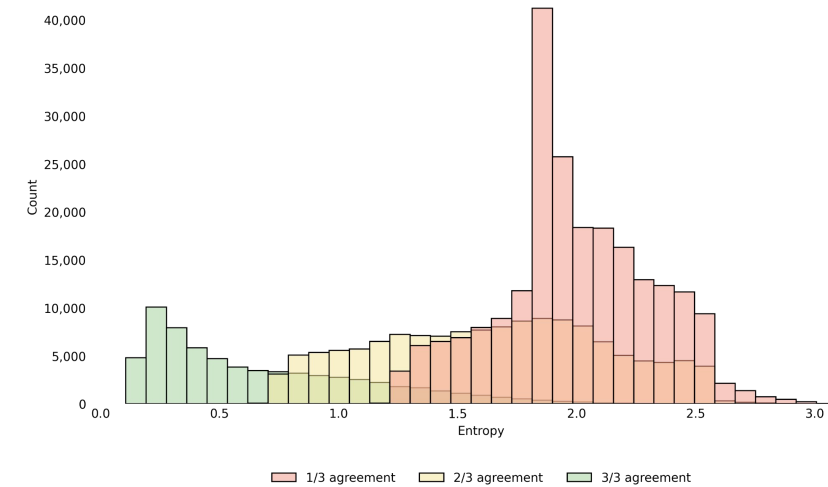


Slovenia

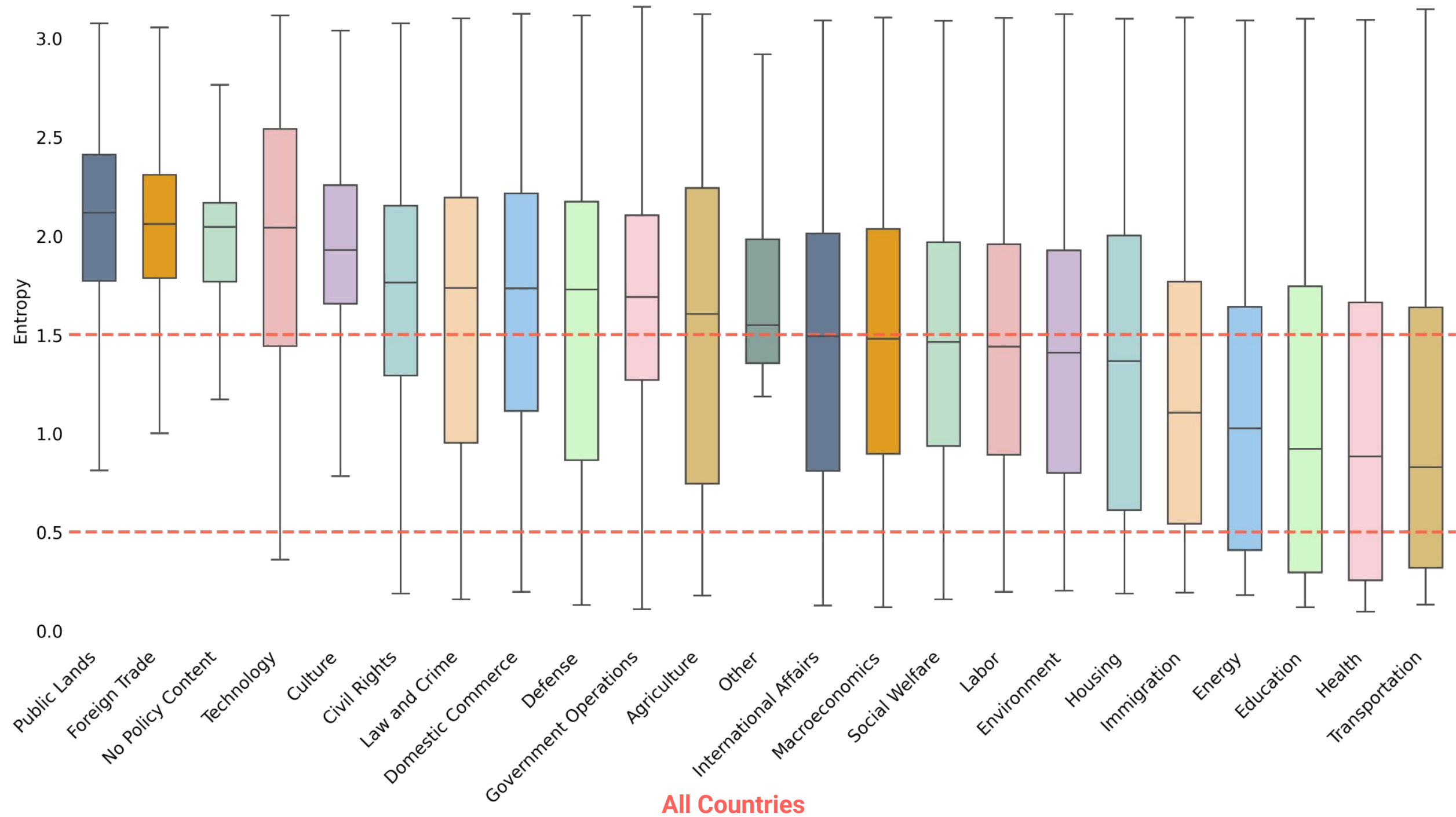
Entropy Distribution



Turkey



Ukraine



All Countries

Benchmark Typology

Gold, Silver, and Bronze

1

Gold

Human-validated

2

Silver

CWCD labels, entropy < 0.5 ; lower-uncertainty benchmark layer

3

Bronze

CWCD labels, entropy 0.5–1.5; higher residual uncertainty, diagnostic layer for future validation

This typology treats benchmark data as a **quality continuum** rather than a binary distinction between automated labels and ground truth. Bronze cases may help identify where additional human validation could promote observations into a more reliable pool.

Entropy-based Retained Pools

Unfiltered Universe

Comprises all observations with non-missing entropy, estimated using the CWCD methodology.

Filtered Universe

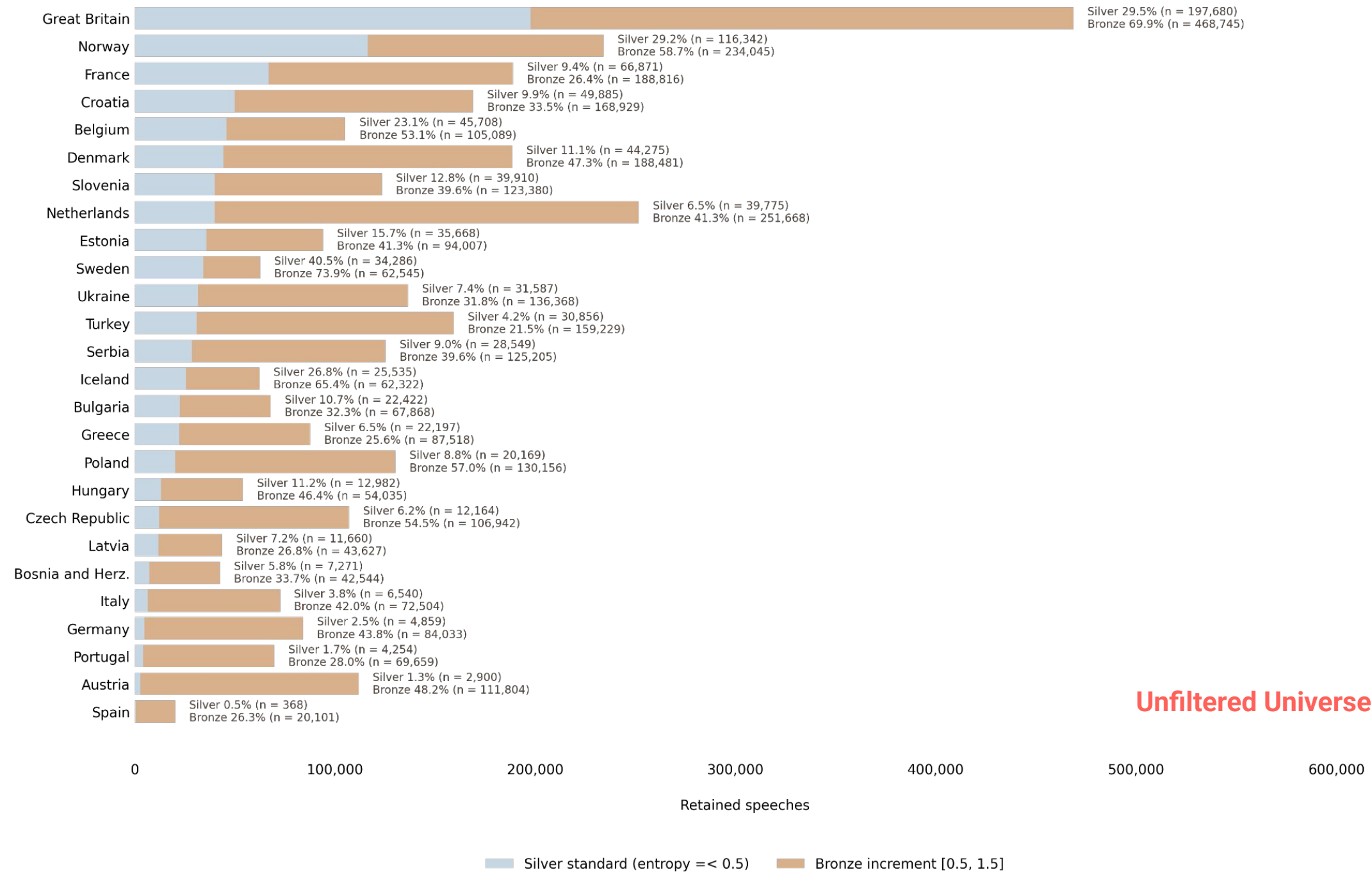
Restricted to **regular target speakers**, crucial for cross-country comparability in model training.

Thresholds & Splits

A **silver threshold of 0.5** and a **bronze threshold of 1.5** define quality. Downstream splits are projected as **70/15/15** (train/validation/test).

Filtering Logic

For ParlaMint countries, the filter requires `speaker_MP == 'MP'` and `speaker_role == 'Regular'`. For Germany (ParlLawSpeech), it excludes speeches by the chair, minister, or chancellor.



Sum of silver retained speeches = 914,713. Average silver retained percentage = 11.6%.
Additional bronze speeches recovered between thresholds 0.5 and 1.5 = 2,344,907.

☆ Standalone & Pooled

Data is robust enough for individual analysis and aggregation. Meets strong thresholds for **retained observations ($\geq 2,000$)**, label imbalance, coverage, and expected test set sparsity.

📦 Pooled Coverage Only

Suitable for pooled use, providing useful policy-label diversity, but not robust enough for standalone splitting. Retains at **least 750 observations** and good label coverage.

📦 Excluded

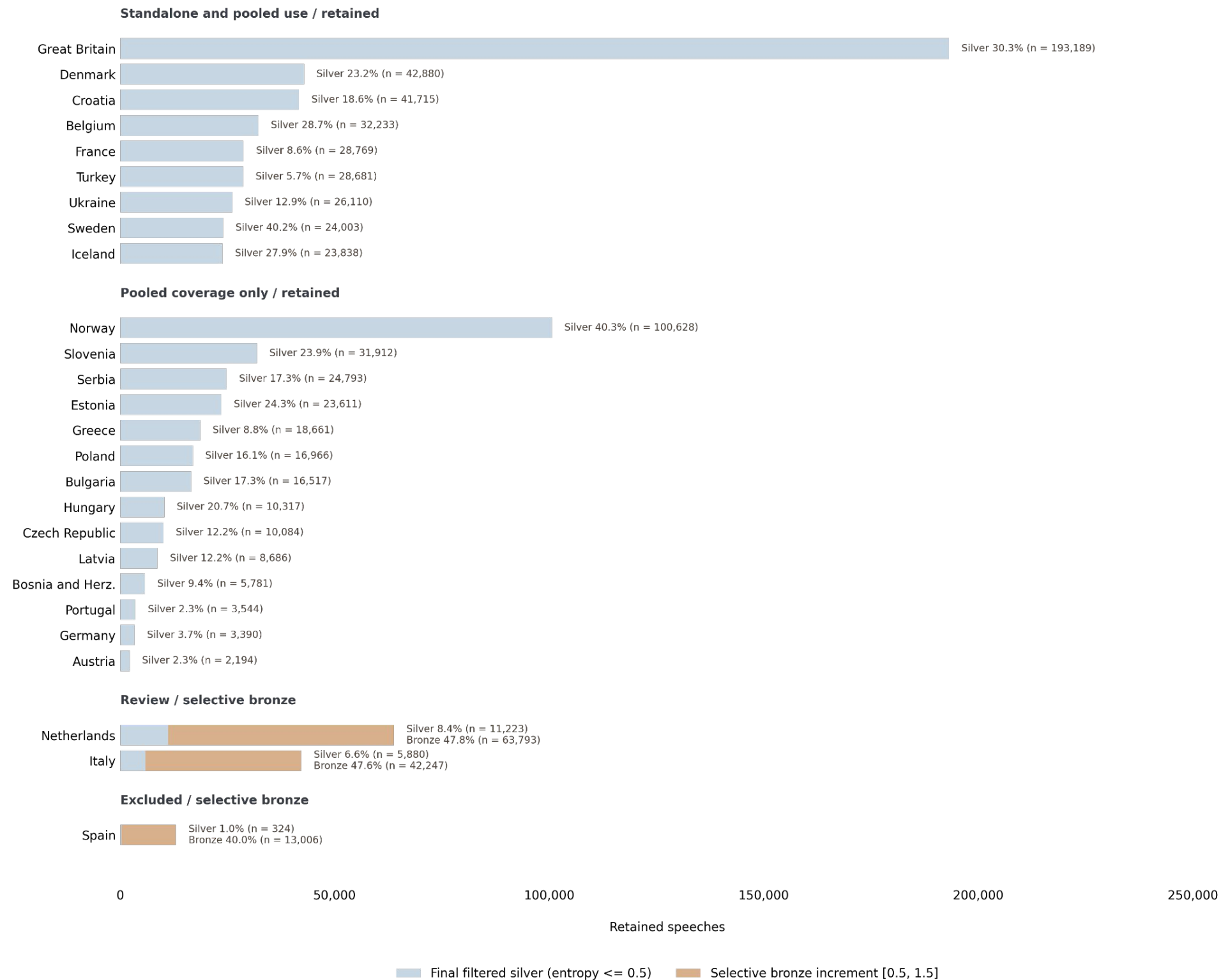
Data falls below minimum thresholds for reliable inclusion at this stage. Indicators like retained observations (< 750), label coverage ($< 25\%$), or **extreme label imbalance**.

🔍 For Review

Does not meet standalone or pooled criteria, nor the exclusion rule. These are mixed cases where one or more **indicators are borderline**, requiring further investigation.

⬆️ Bronze Increment Candidate

Indicates countries where **moving from the silver to bronze** threshold could significantly increase observations, labels, or labels with sufficient data points.



Filtered pools are restricted to MPs with regular speaker roles.
Sum of final pooled silver retained speeches = 718,502. Average pooled silver retained percentage = 17.7%.
Additional bronze speeches shown in the diagnostic plot = 101,619. Bronze increments shown only for review and excluded countries.

Split Strategy

Country-First Splitting

- Each country contributes its own train, validation, and test partitions.
- Pooled train, validation, and test files are concatenations of these country-level partitions.
- No pooled split is created before the country-level split logic is applied.

Stratification

Speeches are split by `predicted_label` (**major topic**) within each country.

Eligible Countries

★ Standalone & Pooled

📄 Pooled Coverage Only

Countries classified as **review** or **excluded** are not included in this release at this stage, focusing on high-quality, actionable data.

OCPSG Silver Standard Dataset

High-Quality Annotations

Comprises parliamentary speeches with **silver standard** policy topic labels, meticulously annotated from the Comparative Agendas Project (CAP) codebook.



Benchmarking Core

Specifically designed to facilitate the rigorous evaluation of both LLMs and fine-tuned models in policy agenda annotation.

[OCPSG-Benchmarking-LLMs/ocpsg-silver-standard-parliamentary-speeches](https://ocpsg-benchmarking-llms/ocpsg-silver-standard-parliamentary-speeches)

Benchmarking LLMs

Once the silver standard is defined, the project proceeds with direct LLM benchmarking under controlled settings.

Zero-Shot

No examples provided; tests raw model knowledge of policy-topic categories.

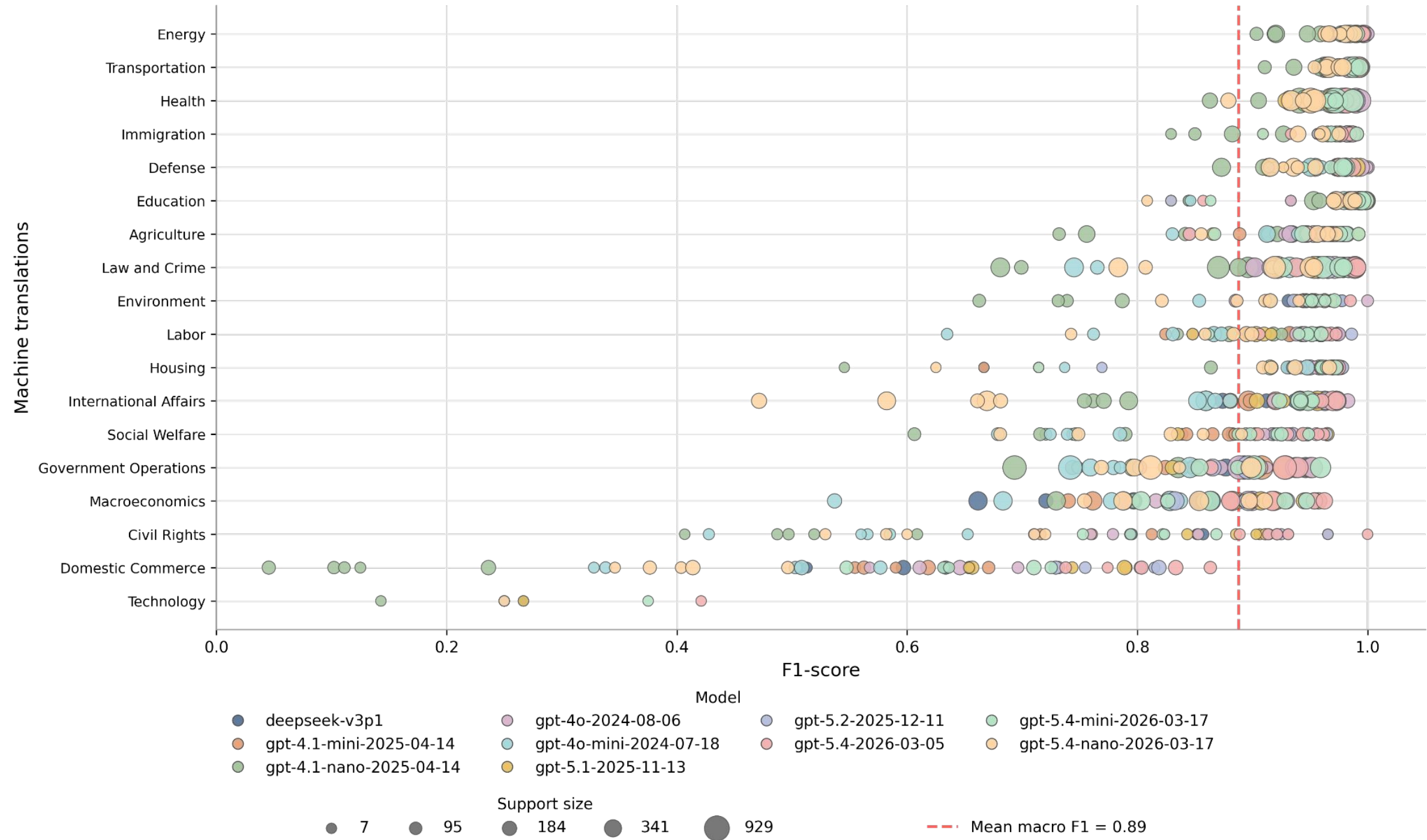
Few-Shot

Exemplars held outside the evaluation partition; tests in-context learning.

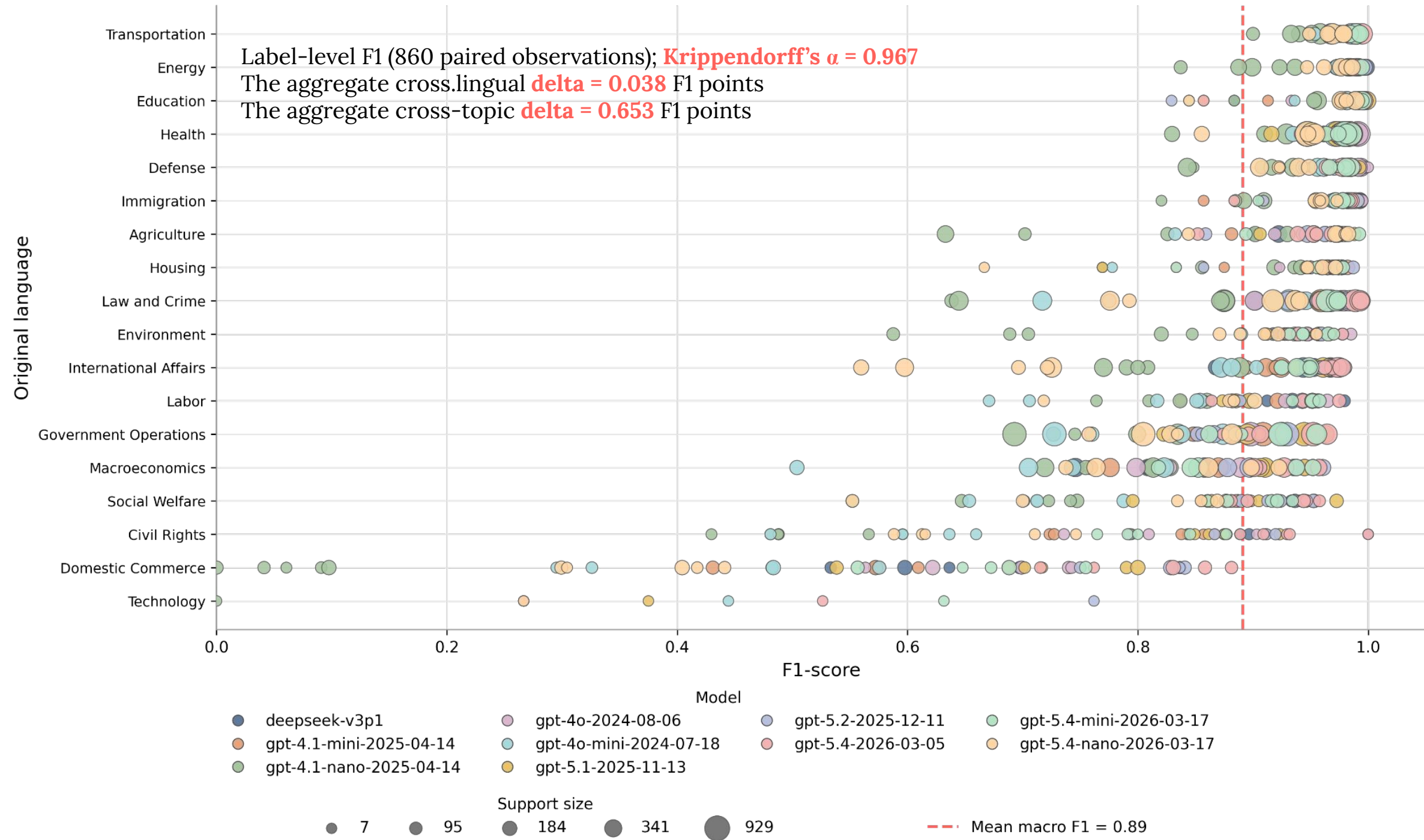
Reasoning-Oriented

Chain-of-thought or structured prompting strategies for complex cases.

Prompt variations are treated as part of the experimental design. Both **deterministic** and **stochastic decoding** regimes are benchmarked explicitly (temperature settings should shape reliability and reproducibility of outputs).



Note. Countries included: Belgium (BE), France (FR), Iceland (IS), Sweden (SE), Ukraine (UA). Total predictions so far: 202,400.
Labels are ordered by mean by-label F1-score across models. The vertical line reports mean macro F1.



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From Bronze to Benchmark

Bastián González-Bustamante, Christopher Klamm, and Tom Bellens

Spanish Language Benchmark

This **spin-off project, funded by Universidad Diego Portales**, aims to develop a validated benchmark extension for Spanish parliamentary speech classification, integrating into the OCPSG framework for multilingual policy annotation.

Current Challenge

Spanish speeches are currently excluded from the **silver-standard dataset** due to imbalance, limiting comprehensive evaluation.

Innovative Approach

We will utilise the larger, more diverse **bronze-standard pool**, transforming moderate-uncertainty labels into reliable data via targeted human validation protocols.

Key Deliverables

A **curated Spanish-focused validated dataset** will be released on Hugging Face, alongside a working paper for submission to a WoS journal.

Rationale for Exclusion: Spain

Silver Standard Exclusion

Spain was excluded from the 'silver-standard' dataset. Only **324 observations** remained, all within a single label with **4.3% coverage** and zero normalised entropy. This made standalone evaluation non-robust.

Bronze Standard Opportunity

Relaxing the criterion to 'bronze-standard' significantly expands the Spanish pool to **13,006 observations across 20 policy domains**. This moderate-uncertainty data is repairable, reframing the exclusion as a research problem.

This spin-off project investigates if uncertain multilingual political text data can be upgraded into valid benchmark material through structured validation and targeted sampling.

Timetable

July - August 2026

Project setup, sampling and annotation protocol, extraction and profiling of Spanish bronze-standard observations, **descriptive analysis by label and entropy**, coder guidelines drafted.

November - December 2026

Comparison of human labels, model disagreement, and entropy; **construction of the validated Spanish-language extension dataset** with cleaning, standardisation, and documentation.

September - October 2026

Pilot annotation, refinement of coding instructions, inter-coder agreement assessment, **main human-validation phase**, and adjudication of difficult cases.

January - March 2027

Hugging Face dataset **release preparation**, drafting and revision of working paper, finalisation of all deliverables, public dataset release, and submission.



Thank you very much!



<https://ocpsg-benchmarking-llms.github.io>